



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
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NACC 'A' Grade Accredited, ISO 9001:2015 Certified

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(NBA Accredited)

Subject- Data Mining and Warehousing (CO314)
Unit –I: Introduction to Data Mining

Prof. S.A.Shivarkar

Assistant Professor

Contact No.8275032712

Email- shivarkarsandipcomp@sanjivani.org.in



Content

- ❖ **Data Mining, Kinds of pattern and technologies**
- ❖ **Data Mining Task Primitives**
- ❖ **Issues in mining**
- ❖ **KDD vs data mining**
- ❖ **OLAP, knowledge**
- ❖ **representation, Information and Knowledge;**
- ❖ **Attribute Types: Nominal, Binary, Ordinal and Numeric attributes, Discrete versus Continuous attributes.**



Why Data Mining

- “We are living in the information age” is popular saying, but actually living in the data age.
- The Explosive Growth of Data: from terabytes to petabytes pour into computer networks by www, various data storage devices everyday from business, society, science and engineering, medicine and almost every other aspect of daily life.
 - Data collection and data availability
 - Automated data collection tools, database systems, Web, computerized society
 - Major sources of abundant data
 - Business: Web, e-commerce, transactions, stocks, ...
 - Science: Remote sensing, bioinformatics, scientific simulation, ...
 - Society and everyone: news, digital cameras, YouTube
- We are drowning in data, but starving for knowledge!
- “Necessity is the mother of invention”—Data mining—Automated analysis of massive data sets



How Data Mining turns large collection of data into knowledge

- A search engine e.g., Google receives hundreds of millions of queries every day can be viewed as a transaction where the user describes her or his information need.
- What novel and useful knowledge can a search engine learn from such a huge collection of queries collected from users over time?
- Interestingly, some patterns found in user search queries can disclose invaluable knowledge that cannot be obtained by reading individual data items alone. For example, Google's Flu Trends uses specific search terms as indicators of flu activity.
- It found a close relationship between the number of people who search for flu-related information and the number of people who actually have flu symptoms.
- A pattern emerges when all of the search queries related to flu are aggregated.
- Using aggregated Google search data, Flu Trends can estimate flu activity up to two weeks faster than traditional systems can.
- This example shows how data mining can turn a large collection of data into knowledge that can help meet a current global challenge.



Evolution of Sciences

- Before 1600, **empirical science**
- 1600-1950s, **theoretical science**
 - Each discipline has grown a *theoretical* component. Theoretical models often motivate experiments and generalize our understanding.
- 1950s-1990s, **computational science**
 - Over the last 50 years, most disciplines have grown a third, *computational* branch (e.g. empirical, theoretical, and computational ecology, or physics, or linguistics.)
 - Computational Science traditionally meant simulation. It grew out of our inability to find closed-form solutions for complex mathematical models.
- 1990-now, **data science**
 - The flood of data from new scientific instruments and simulations
 - The ability to economically store and manage petabytes of data online
 - The Internet and computing Grid that makes all these archives universally accessible
 - Scientific info. management, acquisition, organization, query, and visualization tasks scale almost linearly with data volumes. **Data mining** is a major new challenge!
- Jim Gray and Alex Szalay, *The World Wide Telescope: An Archetype for Online Science*, Comm. ACM, 45(11): 50-54, Nov. 2002

The world is data rich but information poor

- The abundance of data, coupled with the need for powerful data analysis tools, has been described as a data rich but information poor situation
- The data collected in large data repositories become “data tombs”
- Consequently, important decisions are often made based not on the information-rich data stored in data repositories but rather on a decision maker’s intuition, simply because the decision maker does not have the tools to extract the valuable knowledge embedded in the vast amounts of data.
- Efforts have been made to develop expert system and knowledge-based technologies, which typically rely on users or domain experts to manually input knowledge into knowledge bases.
- Unfortunately, however, the manual knowledge input procedure is prone to biases and errors and is extremely costly and time consuming.
- **The widening gap between data and information calls for the systematic development of data mining tools that can turn data tombs into “golden nuggets” of knowledge.**





What is Data Mining

- Data mining (knowledge discovery from data)
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
 - Data mining: a misnomer?
- Alternative names
 - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Watch out: Is everything “data mining”?
 - Simple search and query processing
 - (Deductive) expert systems



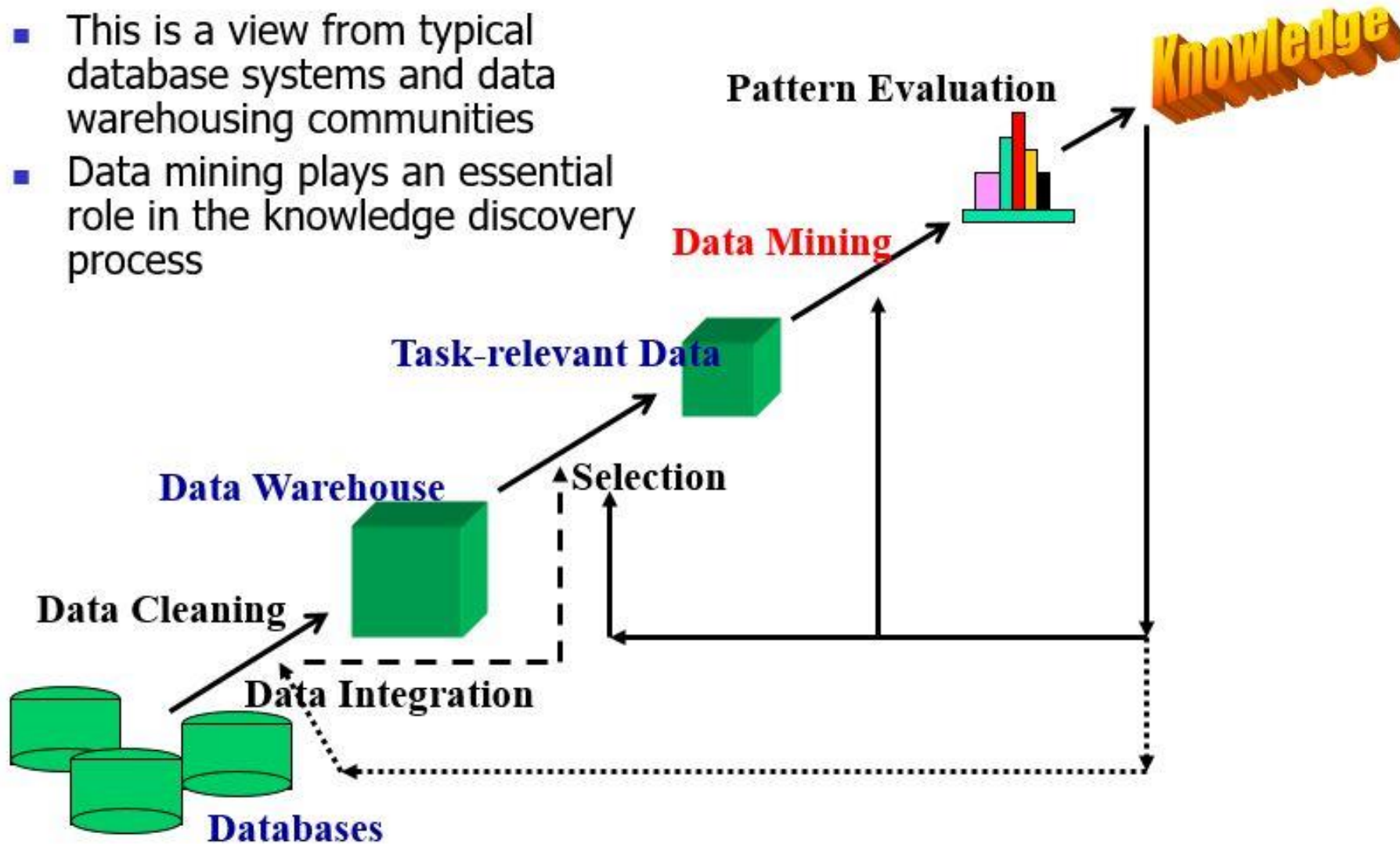
Data mining—searching for knowledge (interesting patterns) in data.

- Many other terms have a similar meaning to data mining—for example, knowledge mining from data, knowledge extraction, data/pattern analysis, data archaeology, and data dredging



Knowledge Discovery (KDD) Process

- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process





Data mining as a step in the process of knowledge discovery

- 1. Data cleaning** : to remove noise and inconsistent data
- 2. Data integration** : where multiple data sources may be combined
- 3. Data selection** : where data relevant to the analysis task are retrieved from the database
- 4. Data transformation** : where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations
- 5. Data mining** : an essential process where intelligent methods are applied to extract data patterns
- 6. Pattern evaluation** : to identify the truly interesting patterns representing knowledge based on *interestingness measures*.
- 7. Knowledge presentation** : where visualization and knowledge representation techniques are used to present mined knowledge to users

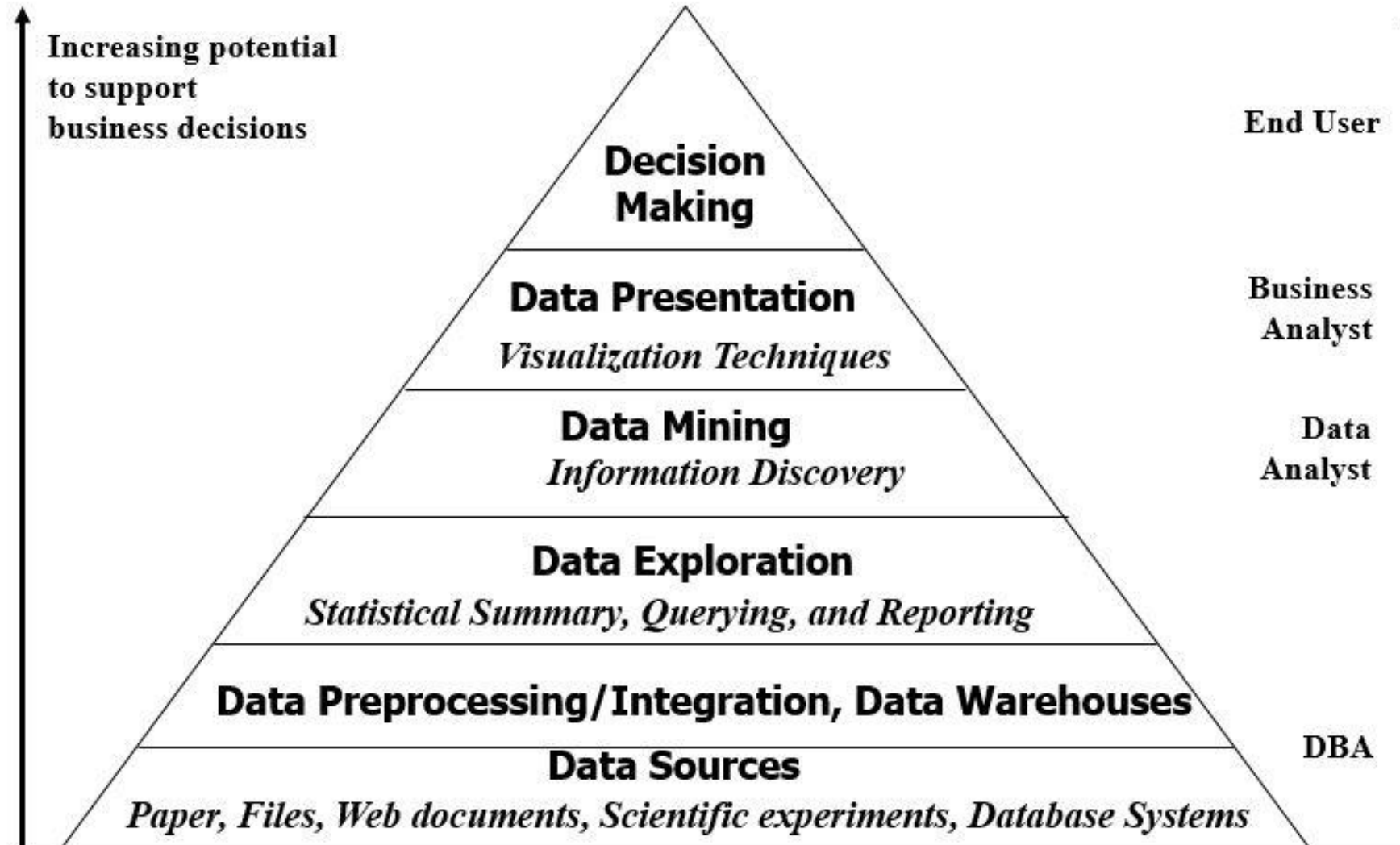


Example: A Web Mining Framework

- Web mining usually involves
 - Data cleaning
 - Data integration from multiple sources
 - Warehousing the data
 - Data cube construction
 - Data selection for data mining
 - Data mining
 - Presentation of the mining results
 - Patterns and knowledge to be used or stored into knowledge-base



Data Mining Business Intelligence (BI)



What is data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe **Objects** an object
 - Object is also known as record, point, case, sample, entity, or instance

Attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Attributes types in Data Mining

- The attribute is the property of the object. The attribute represents different features of the object. Example:
- In this example, Roll No, Name, and Result are attributes of the object student.

RollNo	Name	Result
1	Ali	Pass
2	Akram	Fail



Attributes types in Data Mining

- Binary
- Nominal
- Numeric
- Interval-scaled: e.g. calendar dates, temperature in Celsius or Fahrenheit
- Ratio-scaled: these are like free numbers they can be anything e.g. length, time, counts



Attributes types in data mining conti...

■ Binary:

Binary data have only two values/states

Binary attributes types:

- ☐ Symmetric
- ☐ Asymmetric

☐ Symmetric

Both values are equally important

☐ Asymmetric

Both values are not equally important

Attribute	Value
Cancer detected	Yes, No
Result	Pass, Fail

	Value
	Male, Female

Attribute	Value
Cancer detected	Yes, No
Result	Pass, Fail



Attributes types in data mining conti...

■ Ordinal:

All Values have a meaningful order.

e.g. rankings (scaling from 1 to 10), grades, height (tall, medium, short)

Attribute	Value
Grade	A, B, C, D, F
BPS- Basic pay scale	16, 17, 18

■ Discrete:

Discrete data have finite value. It can be in numerical form and can also be in categorical form.

e.g. number of birds in a flock; the number of heads realized when a coin is flipped 10 times

Attribute	Value
Profession	Teacher, Business Man, Peon etc
Postal Code	42200, 42300 etc



Attributes types in data mining conti...

■ Continuous:

Continuous data technically have an infinite number of steps.

Continuous data is in float type.

There can be many numbers in between 1 and 2.
e.g. weights and heights of birds, temperature of a day

Attribute	Value
Height	5.4..., 6.5..... etc
Weight	50.09.... etc



Attributes types summary

Attribute Type	Description	Examples	Operations
Nominal	The values of a nominal attribute are just different names, i.e., nominal attributes provide only enough information to distinguish one object from another. ($=$, $\neq$)	zip codes, employee ID numbers, eye color, sex: { <i>male</i> , <i>female</i> }	mode, entropy, contingency correlation, χ^2 test
Ordinal	The values of an ordinal attribute provide enough information to order objects ($<$, $>$).	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. ($+$, $-$)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and ratios are meaningful. ($*$, $/$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation



Data Mining Function: (1) Generalization

- Information integration and data warehouse construction
 - Data cleaning, transformation, integration, and multidimensional data model
- Data cube technology
 - Scalable methods for computing (i.e., materializing) multidimensional aggregates
 - OLAP (online analytical processing)
- Multidimensional concept description: Characterization and discrimination
 - Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet region



Data Mining Function: (2) Association and Correlation Analysis

- Frequent patterns (or frequent itemsets)
 - What items are frequently purchased together in your Walmart?
- Association, correlation vs. causality
 - A typical association rule
Diaper $\rightarrow$ Beer [0.5%, 75%] (support, confidence)
 - Are strongly associated items also strongly correlated?
- How to mine such patterns and rules efficiently in large datasets?
- How to use such patterns for classification, clustering, and other applications?



Data Mining Function: (3) Classification

- Classification and label prediction
 - Construct models (functions) based on some training examples
 - Describe and distinguish classes or concepts for future prediction
E.g., classify countries based on (climate), or classify cars based on (gas mileage)
 - Predict some unknown class labels
- Typical methods
 - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, ...
- Typical applications:
 - Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, ...



Data Mining Function: (4) Cluster Analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications

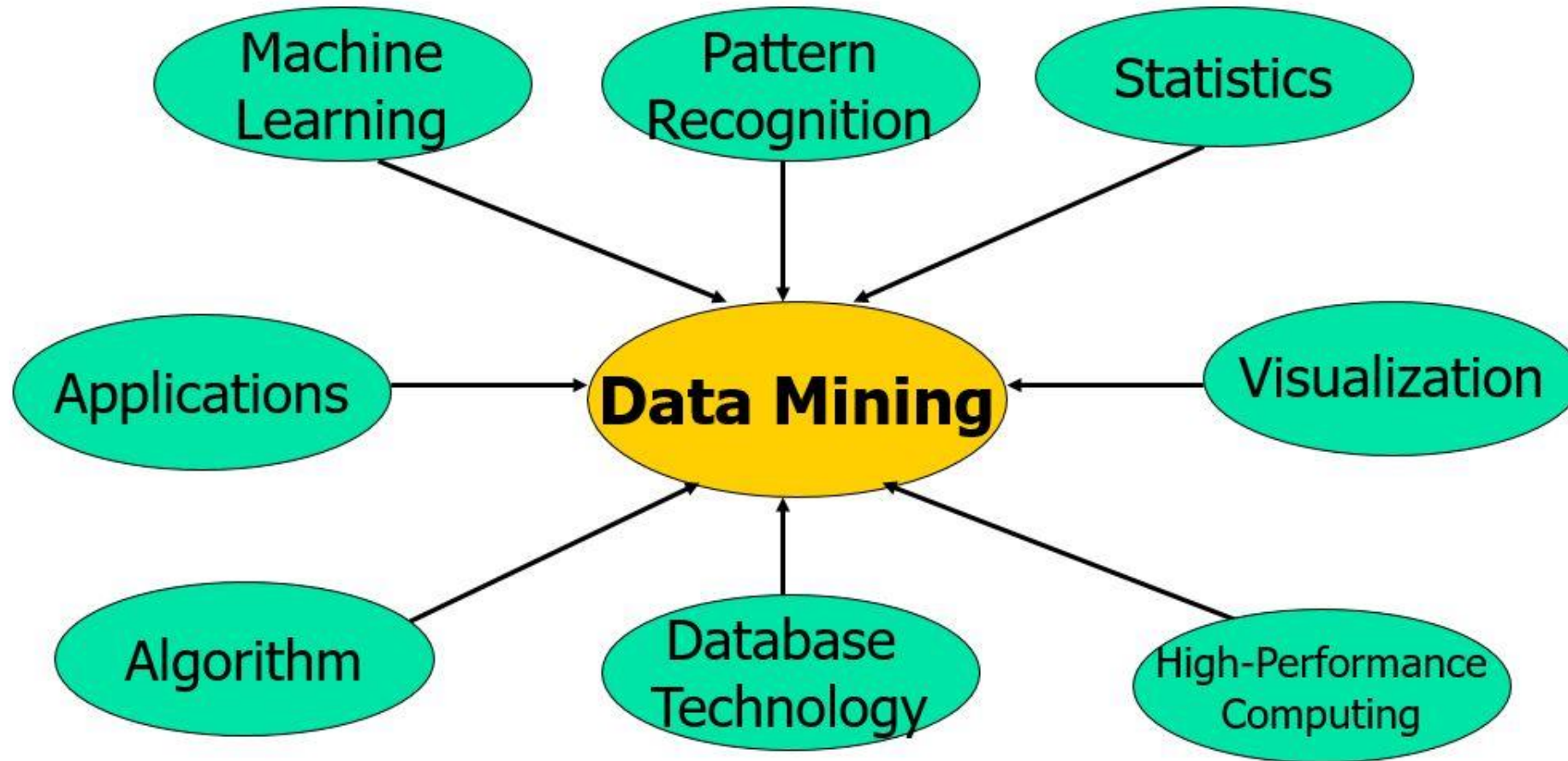


Data Mining Function: (5) Outlier Analysis

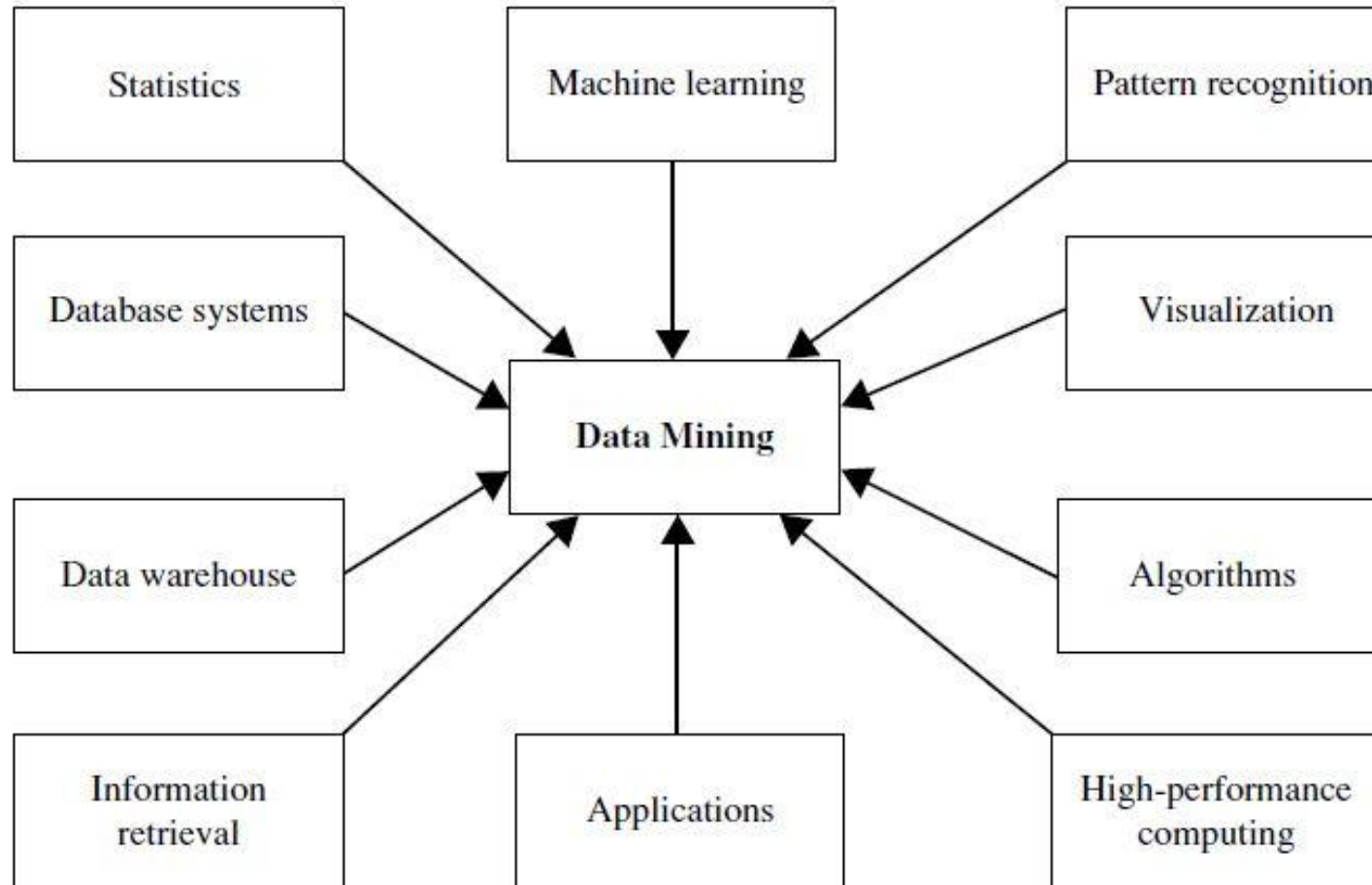
- Outlier analysis
 - Outlier: A data object that does not comply with the general behavior of the data
 - Noise or exception? — One person's garbage could be another person's treasure
 - Methods: by product of clustering or regression analysis, ...
 - Useful in fraud detection, rare events analysis



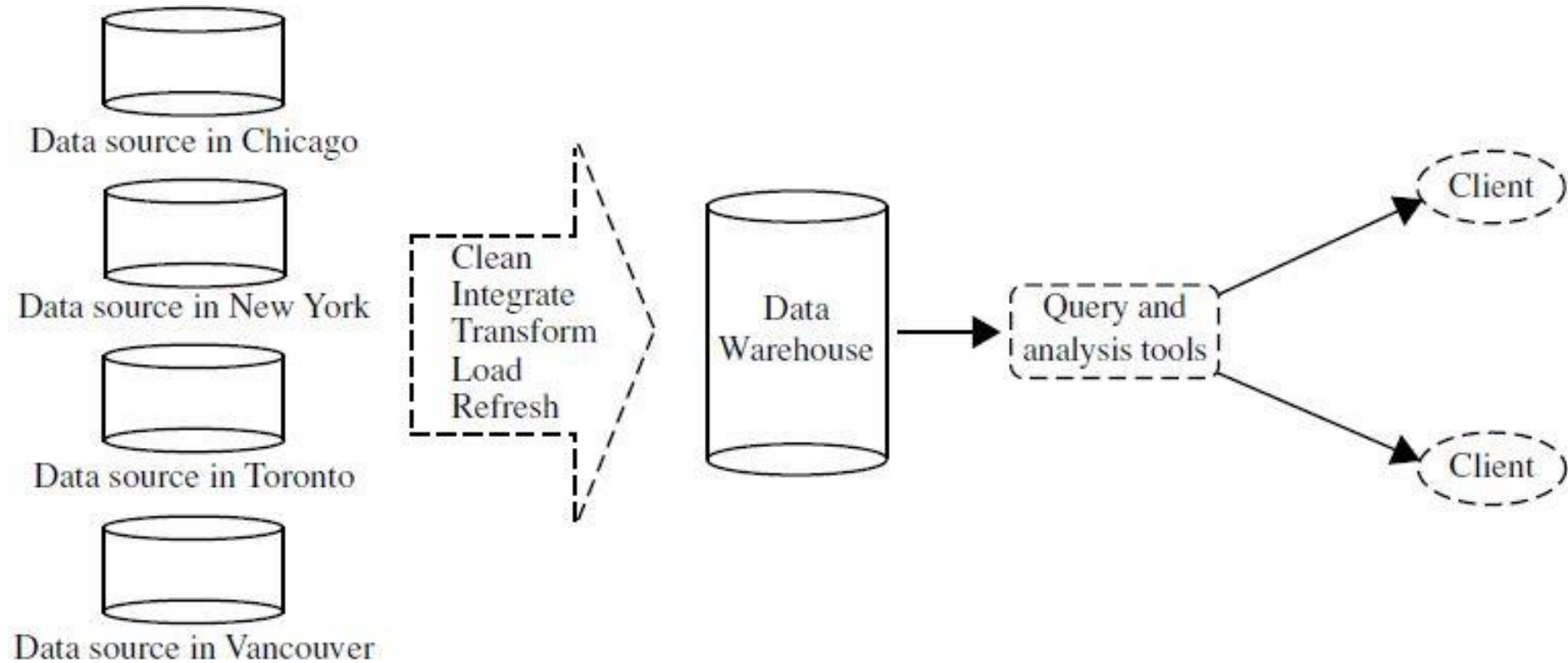
Data Mining: Confluence of Multiple Disciplines



Which Technologies Are Used?

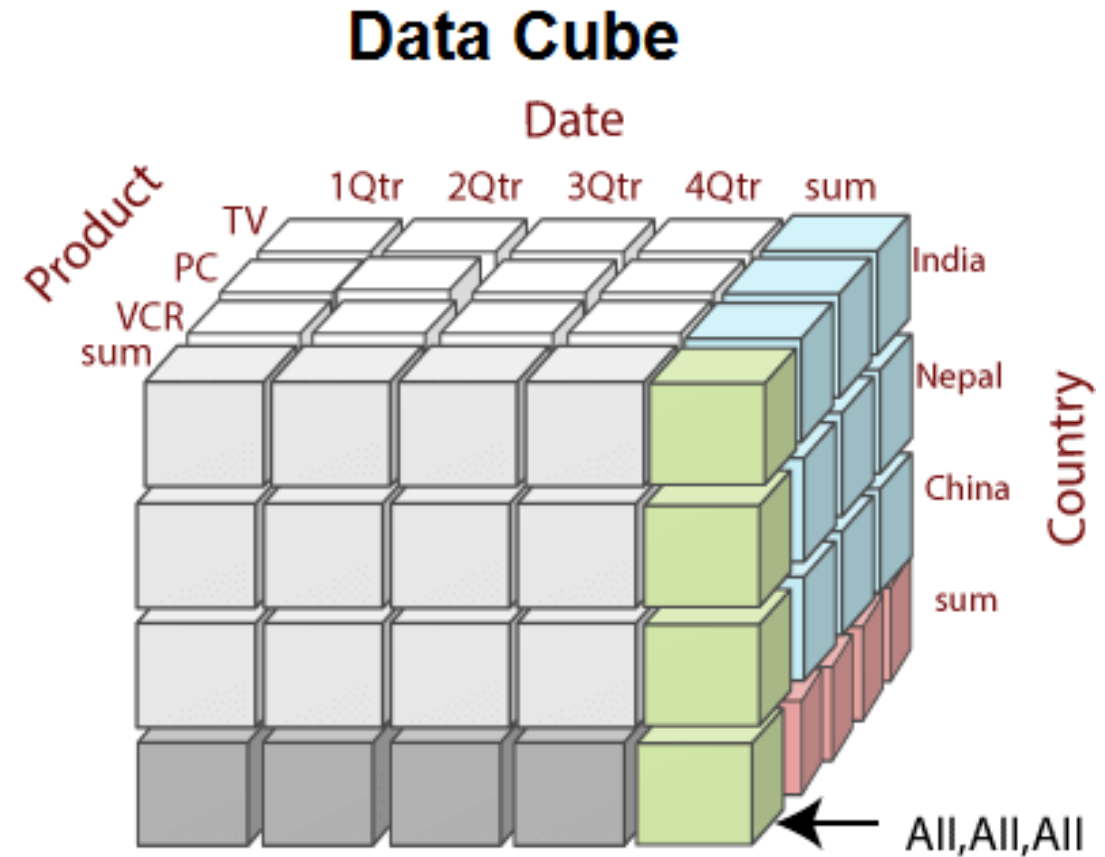


Typical framework of a data warehouse for AllElectronics



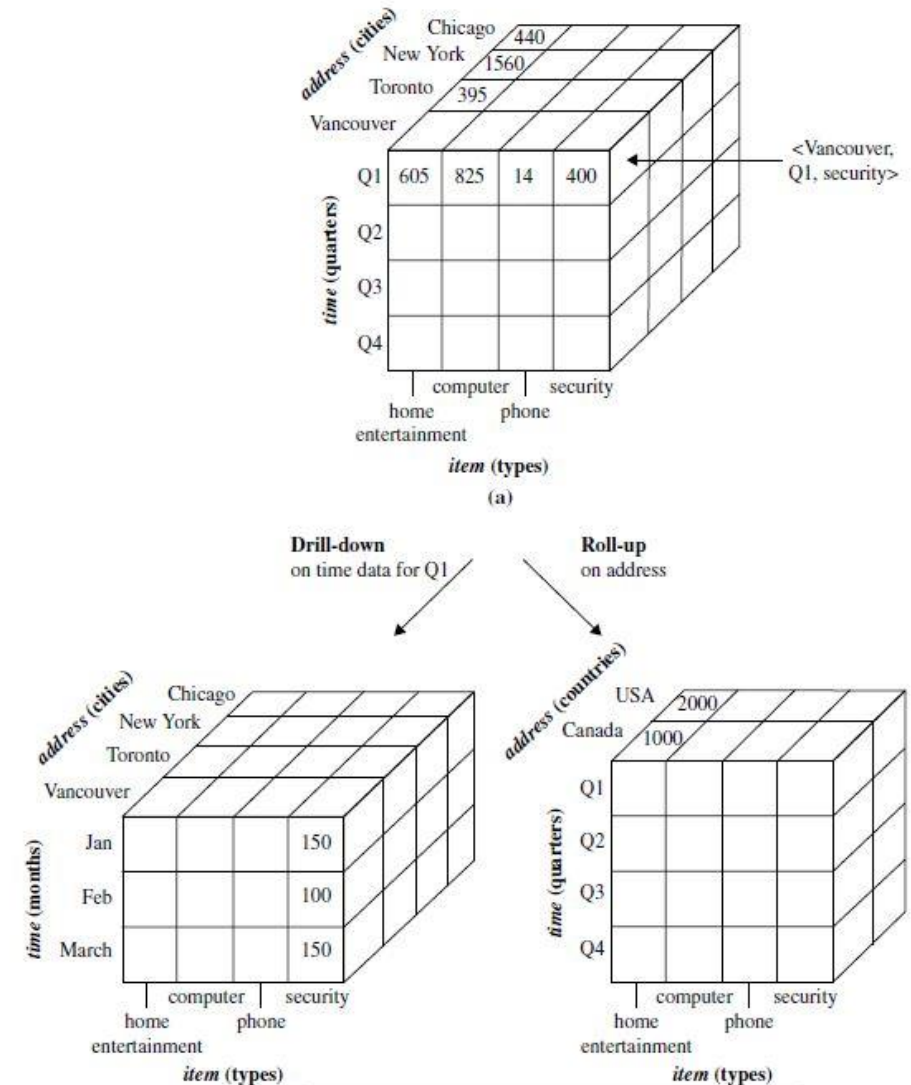
Data Cubes

- Grouping of data in a multidimensional matrix is called data cubes.
- In dataware housing, we generally deal with various multidimensional data models as the data will be represented by multiple dimensions and multiple attributes.
- This multidimensional data is represented in the data cube as the cube represents a high-dimensional space.
- The Data cube pictorially shows how different attributes of data are arranged in the data model.



Typical framework of a data warehouse for AllElectronics

- A Figure multidimensional data cube, commonly used for data warehousing,
- (a) showing summarized data for AllElectronics and
- (b) showing summarized data resulting from drill-down and roll-up operations on the cube in
- (c). For improved readability, only some of the cube cell values are shown.





Reference

- ❖ Han, Jiawei Kamber, Micheline Pei and Jian, “Data Mining: Concepts and Techniques”, Elsevier Publishers, ISBN:9780123814791, 9780123814807.
- ❖ https://onlinecourses.nptel.ac.in/noc24_cs22